



Computer Networks

DS360

Project-Phase 1

Deadline: Day 26/07/2026 @ 23:59

[Total Mark is 5]

Student Details:

CRN:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
 - Phase 1 (Mid project Progress Report)
 - Phase 2 (Final project Report)
- This report is for Phase 1. In this report, you should complete all the requirements for Phase 1 only.
- Each Report will be evaluated according to the marking criteria

*Learning
Outcome(s):*

*Demonstrate
protocol
configuration,
network-
addressing
schemes and
analyze packet
transmission for
the Local and
Wide Area
Networks.*

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. WAN 1 (Main Campus):

- Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
- Connect each LAN to a router (Router 1).

2. WAN 2 (Branch Campus):

- Contains LAN 3 (Library) and LAN 4 (Faculty Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
- Connect each LAN to a router (Router 2).

3. WAN Connection:

- Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
- Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. Assign IP Addresses:

- Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
 - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).
 - LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).

- LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. Router Configuration:

- Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
 - Router 1:
 - LAN 1 Interface: 172.16.10.1/24
 - LAN 2 Interface: 172.16.20.1/24
 - WAN Interface (Serial): 10.10.10.1/30
 - Router 2:
 - LAN 3 Interface: 172.16.30.1/24
 - LAN 4 Interface: 172.16.40.1/24
 - WAN Interface (Serial): 10.10.10.2/30

3. Enable Routing:

- Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.

Phase 1 Requirements**(5 Marks)**

Students are required to submit a short mid progress report, which must include the following:

1. **Work completed this period (2 marks):** This includes the installation of Cisco Packet Tracer, and a study of Packet Tracer for creating nodes, using different types of links (copper straight-through, copper cross-over, serial DCE/DTE), and assigning IP addresses.
 - **Note:** You must show a screenshot of the installation of Packet Tracer (with your system visible, e.g., date/time or username). (1 mark)
 - Give a description of the objectives and advantages of using Packet Tracer for this project. (1 mark)
 2. **One key decision made during this period (1 mark):** For example, your choice of IP addressing scheme, choice of routing method to study, or division of work among group members. Explain briefly why the decision was made.
 3. **One problem encountered (1 mark):** Describe one genuine problem faced so far (technical or organizational) and its current status (resolved / in progress).
 4. **Plan for the next period (0.5 mark):** Briefly list the tasks you plan to complete before the final submission (e.g., building the topology, configuring routers, testing with ping).
 5. **Each group member's input/participation (0.5 mark):** State each member's name and their specific contribution during this period.
-

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

- Live report document (with edit history): [link]
- GitHub/GitLab repository (with commit history): [link]
- Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The mid progress report should be concise including screenshots.
- The same shared document link and repository submitted in Phase 1 must continue to be used for Phase 2, so that the full project history is visible in one place.

Answer:

IP Addressing Scheme

Computer Lab

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/0	172.16.10.1	255.255.255.0	—
Lab-PC1	FastEthernet0	172.16.10.11	255.255.255.0	172.16.10.1
Lab-PC2	FastEthernet0	172.16.10.12	255.255.255.0	172.16.10.1
Lab-PC3	FastEthernet0	172.16.10.13	255.255.255.0	172.16.10.1

Admin Office

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1-Main	GigabitEthernet0/1	172.16.20.1	255.255.255.0	—
Admin-PC1	FastEthernet0	172.16.20.11	255.255.255.0	172.16.20.1
Admin-PC2	FastEthernet0	172.16.20.12	255.255.255.0	172.16.20.1
Admin-PC3	FastEthernet0	172.16.20.13	255.255.255.0	172.16.20.1

Library

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/0	172.16.30.1	255.255.255.0	—

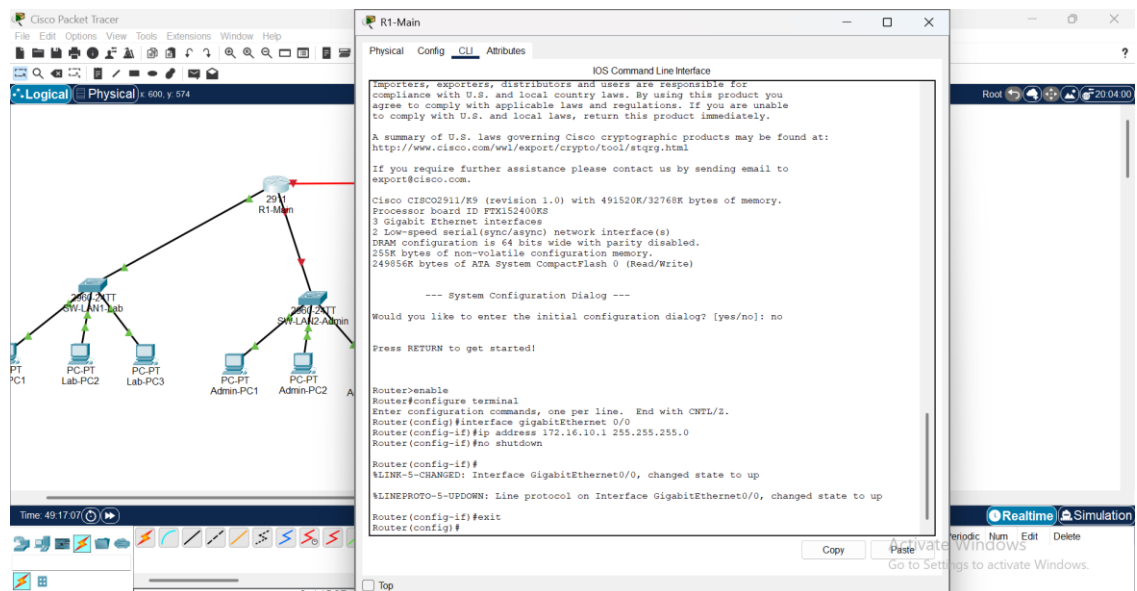
Library-PC1	FastEthernet0	172.16.30.11	255.255.255.0	172.16.30.1
Library-PC2	FastEthernet0	172.16.30.12	255.255.255.0	172.16.30.1
Library-PC3	FastEthernet0	172.16.30.13	255.255.255.0	172.16.30.1

Faculty Office

Device	Interface	IP Address	Subnet Mask	Default Gateway
R2-Branch	GigabitEthernet0/1	172.16.40.1	255.255.255.0	—
Faculty-PC1	FastEthernet0	172.16.40.11	255.255.255.0	172.16.40.1
Faculty-PC2	FastEthernet0	172.16.40.12	255.255.255.0	172.16.40.1
Faculty-PC3	FastEthernet0	172.16.40.13	255.255.255.0	172.16.40.1

WAN Serial Link

Device	Interface	IP Address	Subnet Mask
R1-Main	Serial0/3/0	10.10.10.1	255.255.255.252
R2-Branch	Serial0/3/0	10.10.10.2	255.255.255.252



Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.10.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

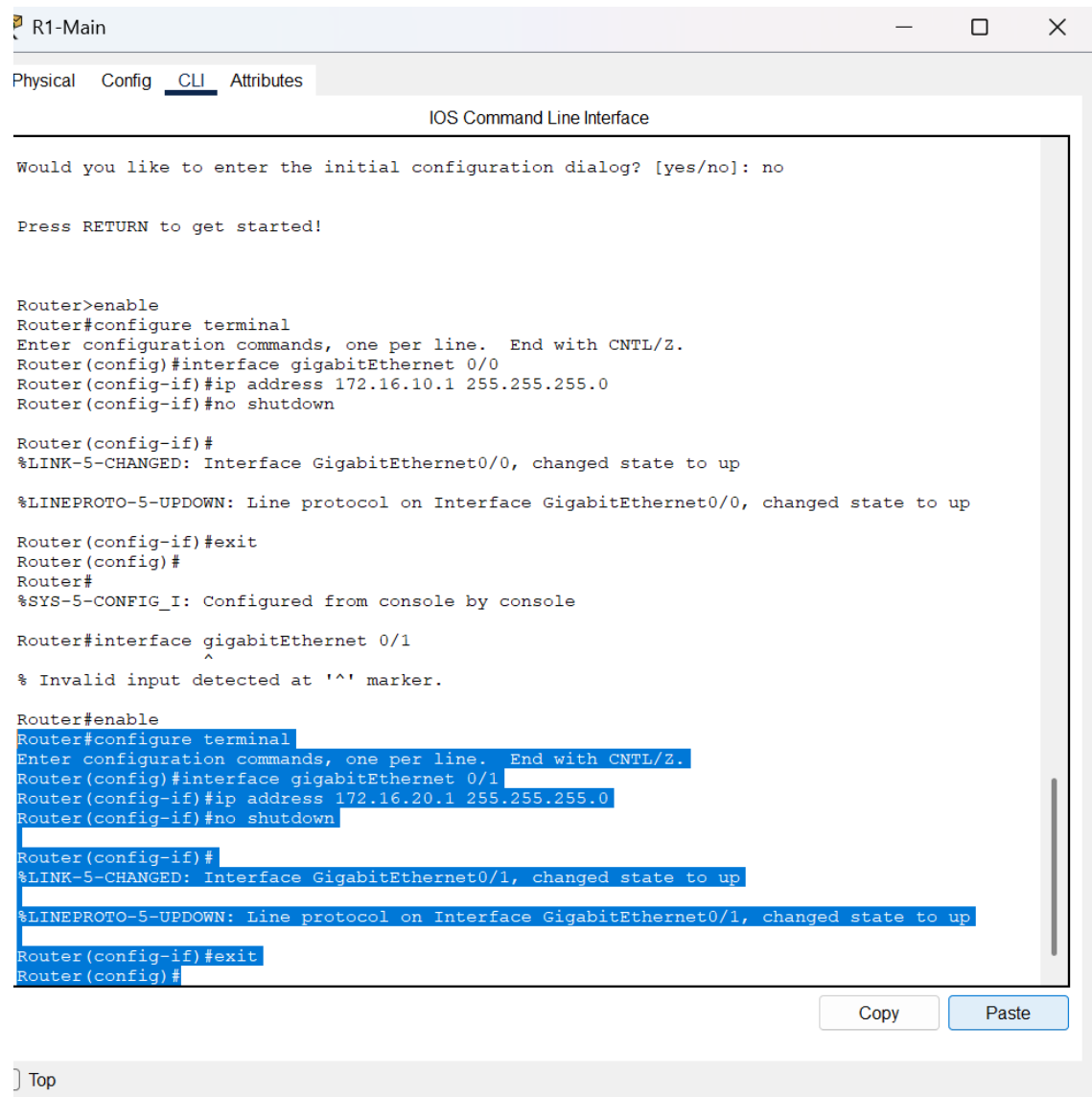
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#

Router#



```
R1-Main
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.16.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#interface gigabitEthernet 0/1
^
% Invalid input detected at '^' marker.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

Copy Paste

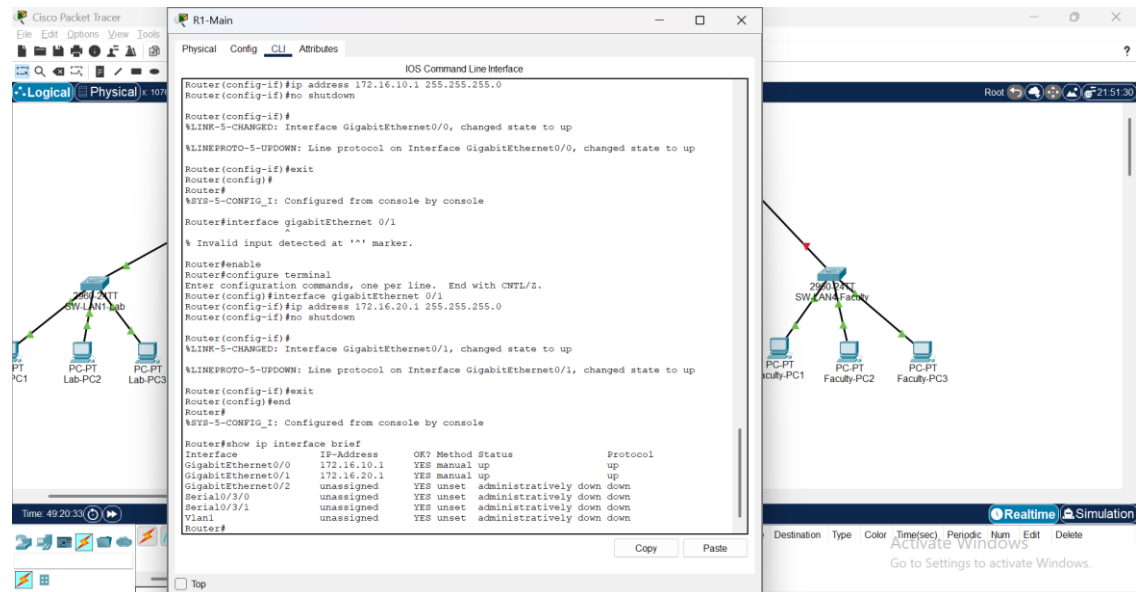
Top

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.20.1 255.255.255.0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

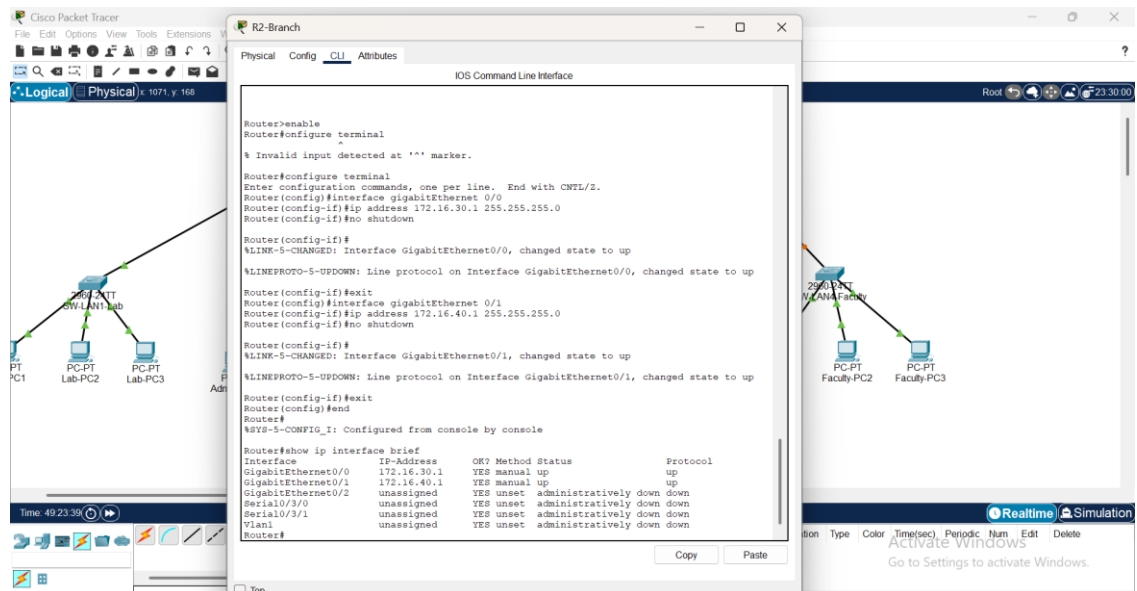
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```
Router(config-if)#exit
Router(config)#
```



```
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.10.1 YES manual up up
GigabitEthernet0/1 172.16.20.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet 0/0

Router(config-if)#ip address 172.16.30.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#interface gigabitEthernet 0/1

Router(config-if)#ip address 172.16.40.1 255.255.255.0

Router(config-if)#no shutdown

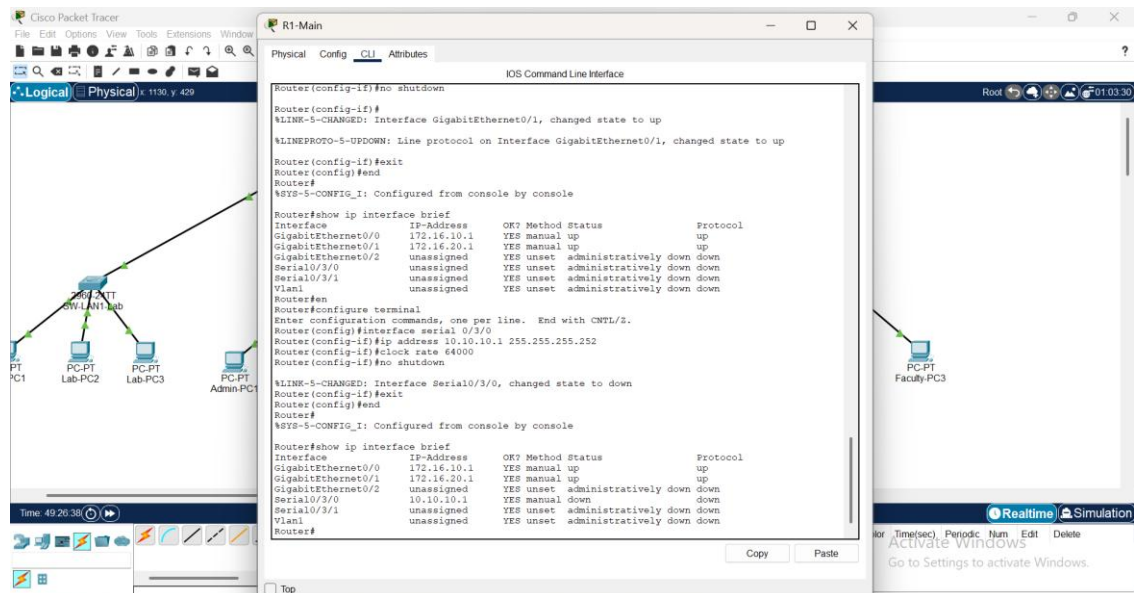
Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 YES manual up up
GigabitEthernet0/1 172.16.40.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
```

```
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.10.1 YES manual up up
GigabitEthernet0/1 172.16.20.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 10.10.10.1 YES manual down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

```

R2-Branch
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       172.16.30.1     YES manual up            up
GigabitEthernet0/1       172.16.40.1     YES manual up            up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/3/0              unassigned      YES unset  administratively down down
Serial0/3/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#end
Translating "end"...domain server (255.255.255.255) % Name lookup aborted
Router#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       172.16.30.1     YES manual up            up
GigabitEthernet0/1       172.16.40.1     YES manual up            up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/3/0              10.10.10.2      YES manual up            up
Serial0/3/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#

```

Copy

☐ Top

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface serial 0/3/0

Router(config-if)#ip address 10.10.10.2 255.255.255.252

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

exit

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#end

Translating "end"...domain server (255.255.255.255) % Name lookup aborted

Router#show ip interface brief

Interface IP-Address OK? Method Status Protocol

GigabitEthernet0/0 172.16.30.1 YES manual up up

GigabitEthernet0/1 172.16.40.1 YES manual up up

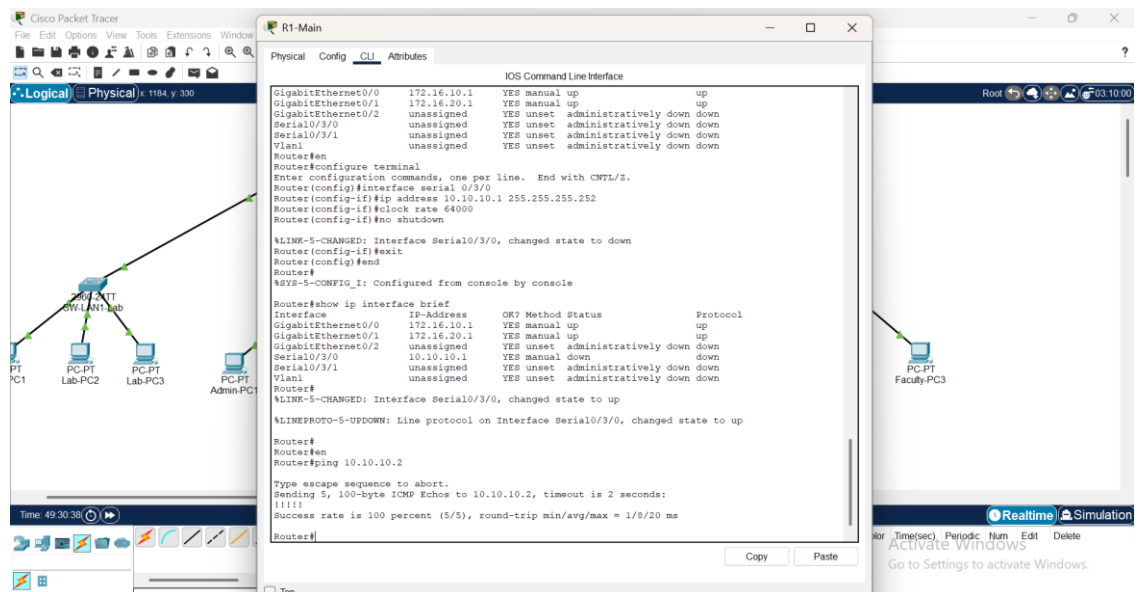
GigabitEthernet0/2 unassigned YES unset administratively down down

Serial0/3/0 10.10.10.2 YES manual up up

Serial0/3/1 unassigned YES unset administratively down down

Vlan1 unassigned YES unset administratively down down

Router#



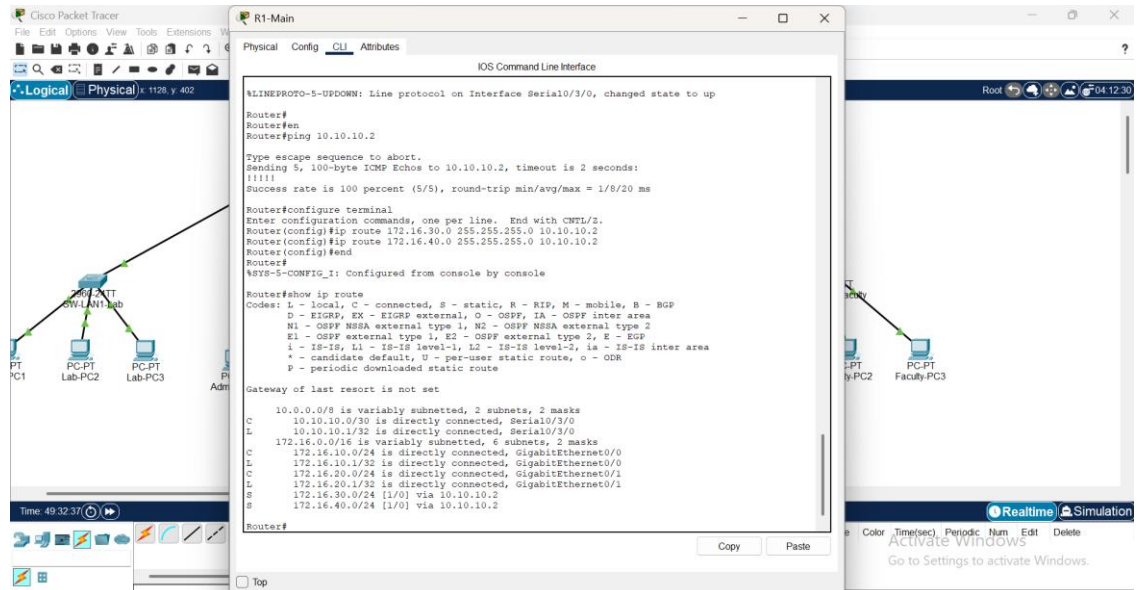
Router#ping 10.10.10.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/20 ms



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 172.16.30.0 255.255.255.0 10.10.10.2

Router(config)#ip route 172.16.40.0 255.255.255.0 10.10.10.2

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.10.10.0/30 is directly connected, Serial0/3/0

L 10.10.10.1/32 is directly connected, Serial0/3/0

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

C 172.16.10.0/24 is directly connected, GigabitEthernet0/0

L 172.16.10.1/32 is directly connected, GigabitEthernet0/0

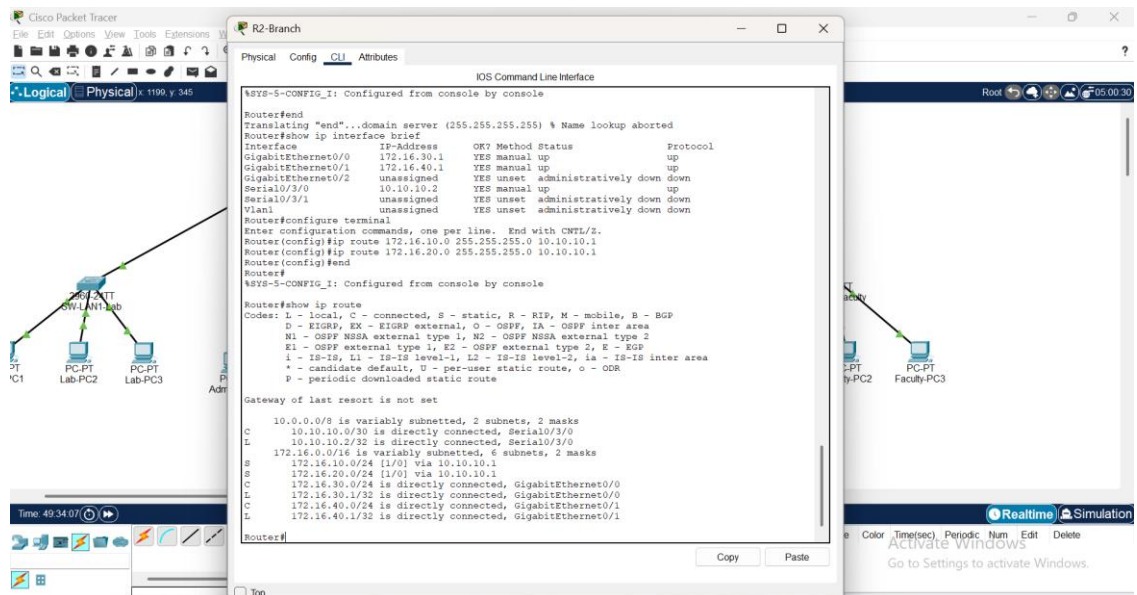
C 172.16.20.0/24 is directly connected, GigabitEthernet0/1

L 172.16.20.1/32 is directly connected, GigabitEthernet0/1

S 172.16.30.0/24 [1/0] via 10.10.10.2

S 172.16.40.0/24 [1/0] via 10.10.10.2

Router#



Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 172.16.10.0 255.255.255.0 10.10.10.1

Router(config)#ip route 172.16.20.0 255.255.255.0 10.10.10.1

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.10.10.0/30 is directly connected, Serial0/3/0

L 10.10.10.2/32 is directly connected, Serial0/3/0

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

S 172.16.10.0/24 [1/0] via 10.10.10.1

S 172.16.20.0/24 [1/0] via 10.10.10.1

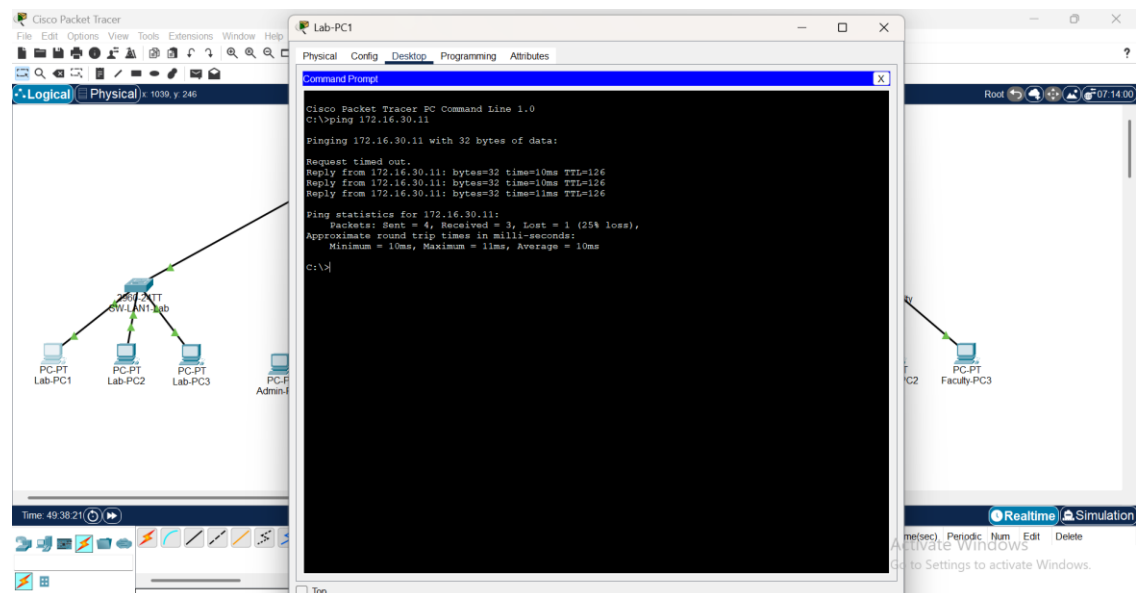
C 172.16.30.0/24 is directly connected, GigabitEthernet0/0

L 172.16.30.1/32 is directly connected, GigabitEthernet0/0

C 172.16.40.0/24 is directly connected, GigabitEthernet0/1

L 172.16.40.1/32 is directly connected, GigabitEthernet0/1

Router#



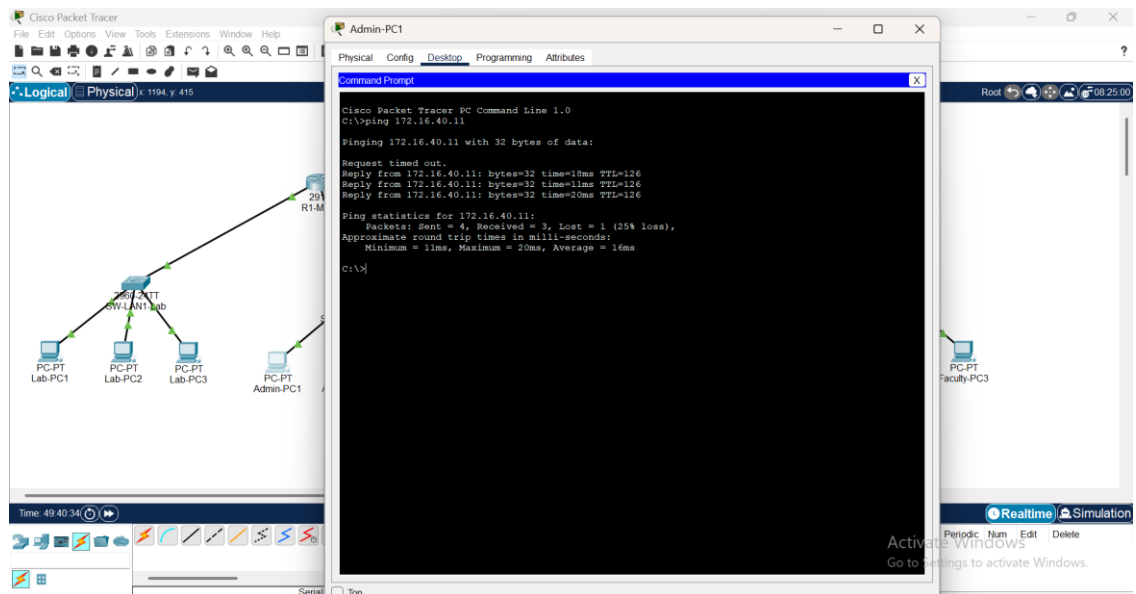
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.30.11

Pinging 172.16.30.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.30.11: bytes=32 time=10ms TTL=126
Reply from 172.16.30.11: bytes=32 time=10ms TTL=126
Reply from 172.16.30.11: bytes=32 time=11ms TTL=126

Ping statistics for 172.16.30.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 11ms, Average = 10ms

C:\>
C:\>
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.40.11

Pinging 172.16.40.11 with 32 bytes of data:

Request timed out.
Reply from 172.16.40.11: bytes=32 time=18ms TTL=126
Reply from 172.16.40.11: bytes=32 time=11ms TTL=126
Reply from 172.16.40.11: bytes=32 time=20ms TTL=126

Ping statistics for 172.16.40.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 20ms, Average = 16ms

C:\>
```

